

# PAPER\_518 — DPM-Unified Inertia, Centripetal, and Centrifugal Forces: Resolving the Classical Conundrum

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**CP4 Class:** DPMUnifiedInertiaCentripetCentrifugCalculator (#113)

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## Abstract

This paper presents a UQFF analysis of DPM-Unified Inertia, Centripetal, and Centrifugal Forces: Resolving the Classical Conundrum, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

## §1 — Abstract

Classical mechanics leaves an unresolved conundrum: inertia is treated as an intrinsic resistance property (Newton's First Law), centripetal force as a real inward constraint force, and centrifugal force as a fictitious pseudo-force appearing only in non-inertial frames. None provides a causative origin for mass. In the Star-Magic UQFF framework all three are **pure forces**, emergent from DPM reaction in 26D layered shells. Their equilibrium defines mass occurrence without invoking any intrinsic property.

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## §2 — Classical Conundrum

| Force                                | Classical Status                     | Problem                                              |
|--------------------------------------|--------------------------------------|------------------------------------------------------|
| Inertia ( $F_{\text{inert}}$ )       | Intrinsic resistance, not a force    | Has no dynamical origin; Mach's Principle unresolved |
| Centripetal ( $F_{\text{centrip}}$ ) | Real inward force (tension, gravity) | Agent-dependent; not fundamental                     |
| Centrifugal ( $F_{\text{centrif}}$ ) | Fictitious pseudo-force              | Vanishes in inertial frame; not a pure force         |

**Result:** Mass cannot be derived from these three; all remain postulates.

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## §3 — DPM-Unified Definitions

### 3.1 — Inertial Force

$$F_{\text{inert}} = -\frac{\partial(DPM_{\text{react}} \cdot \text{ShellEnergy})}{\partial v^{26}} \cdot t_{\text{neg}}$$

- $DPM_{\text{react}}$  is the DPM reaction strength (see PAPER\_516 §3.2)
- $\text{ShellEnergy}$  is the layer-integrated radiance

- $t_{\text{neg}} < 0$ : the negative time factor ensures  $F_{\text{inert}} > 0$  (resists velocity change)
- **Mass occurrence:**  $M = F_{\text{inert}}/a^{26}$

### 3.2 — Centripetal Force

$$F_{\text{centrip}} = \frac{DPM_n(SCm) \cdot \omega_{CW}^2 \cdot r^{\text{layer}}}{1 + \Delta_{\text{dil}}}$$

- DPM north (SCm) component pulls radiance inward (CW direction)
- Dilation correction  $(1 + \Delta_{\text{dil}})^{-1}$ : relativistic shells have weaker centripetal grip (consistent with frame dragging)
- Maintains layered shell coherence; in proto-H shells, condenses smalls into nuclei

### 3.3 — Centrifugal Force

$$F_{\text{centrif}} = DPM_s(UA') \cdot \omega_{CCW}^2 \cdot r^{\text{layer}} \cdot t_{\text{neg}}$$

- DPM south (UA') component pushes radiance outward (CCW direction)
- $t_{\text{neg}} < 0$ : the negative factor makes  $F_{\text{centrif}}$  oppose centripetal (correct sign convention)
- Drives expansion toward Big Bang speed via buoyant radiance outflows

## §4 — Mass Equilibrium Condition

$$F_{\text{inert}} = F_{\text{centrip}} - F_{\text{centrif}}$$

Mass occurs at the point where inertial resistance equals the net shell compression force. The 26D acceleration is:

$$a^{26} = \frac{F_{\text{centrip}} - F_{\text{centrif}}}{M} = \frac{F_{\text{inert}}}{M}$$

**Dual existence symmetry:**

$$F_{\text{centrif,one}} = -F_{\text{centrif,opp}}$$

One-side centrifugal drives expansion; opposite side infers inward compression — the universe self-balances across the 26D egg.

## §5 — Resolution of the Classical Conundrum

| Classical Problem         | Star-Magic Resolution                                                       |
|---------------------------|-----------------------------------------------------------------------------|
| Inertia has no origin     | $F_{\text{inert}}$ emerges from DPM grinding via $\partial/\partial v^{26}$ |
| Centrifugal is fictitious | $F_{\text{centrif}}$ is a real DPM south force (UA' CCW push)               |
| Mass origin unknown       | $M = F_{\text{inert}}/a^{26}$ — mass is equilibrium, not intrinsic          |

| Classical Problem | Star-Magic Resolution                                 |
|-------------------|-------------------------------------------------------|
| Mach's Principle  | Inertia encoded in global DPM layered shell reactions |

The conundrum dissolves as a downward projection artefact: 26D grinding unifies all three, and in the 3D observable slice they appear distinct only because the layered structure is not directly visible.

## §6 — Atomic Mass Uniqueness

Non-repeating quantum fingerprints per atom arise because each atom's  $F_{\text{inert}}$  is set by its specific layer index  $l$ , its local  $(DPM_n, DPM_s)$  reaction state, and the  $t_{\text{neg}}$  dilation at that spacetime location. No two atoms share the same radiance history  $\rightarrow$  unique mass values per atom confirmed by the Standard Model mass spectrum.

## §7 — Canonical Constants

| Symbol               | Value                | Units |
|----------------------|----------------------|-------|
| $DPM_n$ (normalised) | 1.0                  | —     |
| $DPM_s$ (normalised) | 0.85                 | —     |
| $\omega_{CW}$        | $1.2 \times 10^{10}$ | rad/s |
| $\omega_{CCW}$       | $8.3 \times 10^9$    | rad/s |
| $\kappa$             | $5 \times 10^{-4}$   | —     |

## §8 — Conclusion

Inertia, centripetal, and centrifugal forces are unified as pure emergent forces from DPM-layered radiance in 26D shells. Their equilibrium condition  $F_{\text{inert}} = F_{\text{centrip}} - F_{\text{centrif}}$  defines mass occurrence without invoking intrinsic properties. The 3,000-year classical conundrum of the status of centrifugal force is resolved.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.182 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 23, \quad n_{\text{channel}} = 25/26$$

Since  $p_{\text{DVP}} = 23$  is **sub-threshold** (threshold at  $p > 26$ ), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10<sup>4</sup> yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                             | Status                 |
|----------------|----------------------------------------------|----------------------------------------|------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.182                | ✓ Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 23$                  | ✓ Sub-threshold        |
| BSH layers     | 26 harmonic terms                            | $j = 1 \dots 26,$<br>$\cos(2\pi j/26)$ | ✓ Full 26D projection  |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential             | ✓ Canonical            |
| [SSq]          | 0.57                                         | Applied in BSH saturation              | ✓ Canonical            |

### §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable            | UQFF Prediction                                                              | SM / Experiment                    | Source                                        | Alignment                                            |
|-----------------------|------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------|------------------------------------------------------|
| Higgs mass<br>m_H     | UQFF<br>K_HIGGS=47.34 →<br>m_H_UQFF =<br>125.09 GeV                          | m_H = 125.20 ±<br>0.11 GeV         | PDG<br>2024                                   | 99.8%                                                |
| Cosmological<br>Λ     | UQFF                                                                         | ∇UA                                | <sup>2</sup> →<br>1.09e-52<br>m <sup>-2</sup> | Λ =<br>1.114e-52<br>m <sup>-2</sup><br>(Planck+DESI) |
| Thomson σ_T<br>(QED)  | UQFF U_m kernel:<br>σ_T = 6.6524e-29<br>m <sup>2</sup>                       | σ_T = 6.6524e-29<br>m <sup>2</sup> | PDG<br>2024                                   | 100% (exact)                                         |
| κ baryon<br>stability | κ = 0.0005/day;<br>scale separation<br>10 <sup>33</sup> from proton<br>decay | τ_p > 7.7e33 yr<br>(Super-K)       | Super-K<br>2024                               | ✓ UQFF<br>baryon-safe                                |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200$  PeV) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite *PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator)* for full UQFF-SM bridge.

See also: *PAPER\_516 (DPM Shell Radiance)*, *PAPER\_517 (Negative Time Proof)*, *PAPER\_519 (Shell Radiance Prototype)*, *PAPER\_520 (Session 140 Hub)*.

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                    | Key Result                                                        |
|-----------------------------|--------------------------------------------|-------------------------------------------------------------------|
| fneutron_s26_coupling.py    | F_neutron x S_26 buoyancy-polylog coupling | $\sim 470\times$ amplification via 26-level VDS                   |
| kozima_scm_cross_section.py | SCm-modulated neutron-drop cross-section   | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}]^n/26)$ |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)      | FNeutronForce, SigmaSCm, SCmActivation                            |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\text{U,Bi}}/F_{\text{U}} - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2*\Gamma_2)}] * (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                      | Key Result                |
|--------------------------|----------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | S_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)            | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                   | Key Result                  |
|-------------------|-------------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q),<br>psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:** -  $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_n$  -  $\phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n$  -  $\psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

#### S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                    | Key Result                                    |
|------------------------------|--------------------------------------------|-----------------------------------------------|
| ramanujan_pi_uqff.py         | Classical +<br>UQFF-modified 1/pi +<br>26D | 21 digits classical, 15 UQFF, 7<br>digits 26D |
| mock_theta_pi_wstp_kernel.wl | Symbol Wolfram export<br>(UQFFMockThetaPi) | qPochhammer, f26,<br>oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$   
where  $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa_{\text{appai}} n/26)]$

#### S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_SCm     | 0.99                                  | SCm manifold completeness     |
| rho_SCm   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| sigma_0   | $10^{-4}$                             | Base neutron cross-section    |

*Implementation:* all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_1\_CoAnQi.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).